A Prescriptive Analytics Approach to Markdown Pricing for an E-Commerce Retailer

Andrew Vakhutinsky[1], Kresimir Mihic[2] and Su-Ming Wu[3]

[1]Oracle Labs, Burlington, MA
[2]Oracle Labs, Redwood Shores, CA
[3]Oracle Retail Science Group, Burlington, MA
[*]email: firstName.lastName@oracle.com

Abstract

This paper introduces a prescriptive analytics approach to solving markdown-pricing optimization for an e-commerce retailer capable of price differentiation based on customer demand elasticity and the cost of delivery or other services. We consider a situation when the retailer has a limited but potentially large amount of inventory that is stored at multiple fulfillment centers and must be sold by a certain exit date. The objective is to maximize the gross profit, defined as the total revenue minus total shipping cost. We propose a model which predicts, based on historical data, the demand from each customer group as a function of price. Then we formulate the optimization using non-linear objective function and constraints and describe a so-called randomized decomposition approach to finding a near-optimal solution. Finally, we discuss the results of our computational experiments.

1 Introduction

The e-commerce channel of the retail industry is currently experiencing very fast growth. According to the U.S. Census Bureau [USC], in 2017, e-commerce retail sales in the United States grew by 15.5%, whereas the entire retail sector's revenue grew only by 4.3%. At the same time, profit margins have stayed very low. For example, as reported in [Rig14], Amazon.com has averaged only 1.3% in operating margin over the past three years, compared to 6% to 10% for department/discount stores. Therefore, even a small revenue increase that does not involve additional operating cost translates into a significant profit margin increase for an e-commerce retailer.

A typical e-commerce retailer fulfills orders from multiple warehouses, or fulfillment centers (FC), that are generally geographically dispersed and thus the cost of shipping an order to a customer can differ significantly depending on the FC. At the same time, when an item approaches its end of life, its

inventory levels frequently vary greatly among the FCs. Therefore, maximizing total profit should involve both assigning FCs to customers in order to balance the load among the FCs as well as pricing the item based on each customer's price elasticity and cost of service.

In this paper we consider only items with a limited life cycle, such as fashion merchandise or electronics. These items lose nearly their entire value after a certain amount of time. When such an item approaches its end of life, its inventory is usually not replenished. If the amount of inventory is high enough that it cannot all be sold by the exit date at its current price, a markdown is applied. A large body of literature considers different aspects of markdown and related optimization problems. A recent survey of these problems is in [CC15]. Some recent research papers focus on omni-channel retail [HSU18] and optimal inventory levels at FCs [LJS17] as well as markdown with an assortment-dependent demand model, [VKG12] and [SA98]. The novelty of our approach is that it jointly optimizes pricing and FC-to-customer assignment. Thus it integrates supply chain optimization and revenue optimization, the two main areas of retail operations management.

The optimization requires a comprehensive predictive model which accounts for various exogenous and endogenous effects typically confronting the retailer, such as holiday-driven change in consumer demand, presence of similar items resulting in demand transference, a.k.a. demand cannibalization, and pricing and promotion effects. Some existing studies in the area of revenue management describe similar models incorporating such effects, for example [CG12] for the case of a fast-fashion retailer, [FLSL15] for the case of an online fashion retailer, and [IF17] for a grocery retailer. However, these papers model demand by using a relatively small customer group with very sparse sales, leading to very unreliable model parameter estimations. We attempt to address this issue with a hierarchical demand model capable of estimating various model parameters at specific levels.

We propose solving the markdown optimization problem in two stages: In the first stage, we initialize the predictive demand model and then estimate its various model parameters. The model is described in section 2. In the second stage, we find the optimal solution to the prescriptive optimization problem described in section 3, with the optimization performed by the Randomized Decomposition (RD) Solver, a solver designed by one of the authors [MRW18] and described in section 4. Finally, the results of the computational experiments are described in section 5.

2 The Hierarchical demand model

Since many items have either a limited history of sales or low historic price variability, parameter estimates are very unreliable if done using individual items in isolation. On the other hand, estimates based solely on sales aggregated to higher levels of the merchandise hierarchy may introduce a significant bias for certain items. In order to mitigate these two effects, we propose the hierarchical

parameter estimation procedure described below. This approach is similar to ridge regression with non-zero priors, which is closely connected to Bayesian regression.

We propose an iterative solution algorithm alternating between two stages: a bottom-up stage to pool together observations for similar products and a top-down stage to estimate individual product parameters based on the priors obtained from the bottom-up stage.

2.1 The Demand Model

2.1.1 Assumptions

We assume that for each item i there are observations from historic sales data that consist of triples (s_{it}, p_{it}, r_{it}) representing, respectively, sales, price, and a binary promotion flag for time period t. We also assume without loss of generality that the data represent weekly figures. We further assume that during each week the sales are non-zero and there is no inventory shortage, so that sales volume is equal to the demand for the item. In addition, we assume that the merchandise hierarchy is known and adequately reflects the similarity between the retail items. We also assume that the sales history spans at least two years to allow inference of seasonal lifts, particularly those of holiday week sales.

2.1.2 Item-level model

The demand $d_i(t)$ for item i at time t is modeled as:

$$d_i(t) = f_i(p_i(t), r_i(t), b_i(t), t)$$

where $f_i()$ is the functional dependency on the independent variables. If we model the dependence on the time variable t as a multiplicative seasonality effect $s_i(t)$, the model can be rewritten as

$$d_i(t) = s_i(t) f_i^{deseas}(p_i(t), r_i(t), b_i(t))$$

where $f_i^{deseas}()$ expresses the so-called deseasonalized demand.

Since the time variable t is usually expressed in terms of the week of the year and the sales history of a particular item is limited to two years, or about 100 weeks, there is not enough statistical power to estimate $s_i(t)$ for each week. Therefore, it is more practical to express seasonality as the product of demand spikes during certain high demand weeks chosen a-priori, denoted W_{hol}, and the exponentiated sine curve $\exp(\beta_i^{sin} \sin(\frac{2\pi}{52}t) + \beta_i^{cos} \cos(\frac{2\pi}{52}t))$, where β_i^{sin} and β_i^{cos} are estimated.

The model also contains a demand cannibalization variable for each item that quantifies the negative impact on its demand from similar items. This effect is also known as demand transference or substitution. The specifics of computing the cannibalization variables are provided in section 2.1.3.

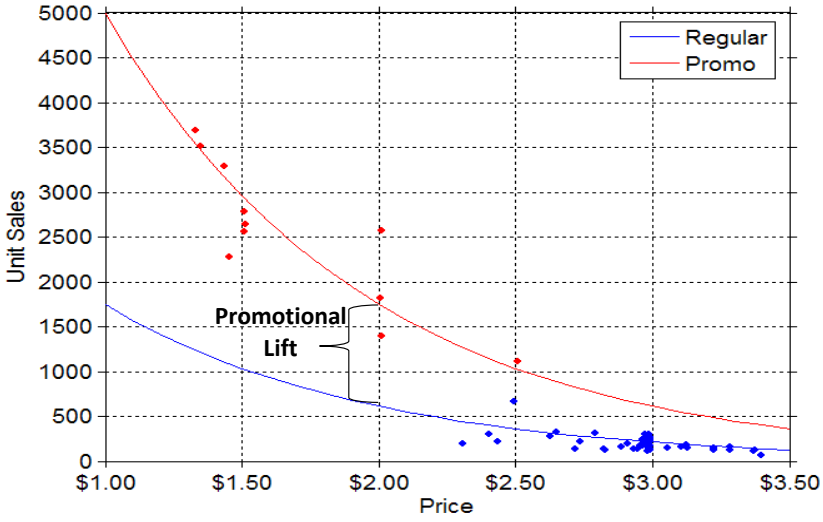


Figure 1: Illustration of fitting the observed sales accounting for promotional lift effect

Thus, assuming log-normal demand, the demand model can be formulated as:

$$\log d_i(t) = \beta_i^0 + \beta_i^{price} p_i(t) + \beta_i^{promo} r_i(t) + \beta_i^{cann} b_i(t) + \beta_{it}^{hol} I_{W_{hol}}(t) + \beta_i^{sin} \sin(\frac{2\pi}{52}t) + \beta_i^{cos} \cos(\frac{2\pi}{52}t) + \varepsilon \quad (1)$$

where $I_{W_{hol}}()$ is an indicator variable for the pre-selected holiday weeks and ε is normally-distributed, zero-mean error term. An example of fitting the deseasonalized model using price sensitivity β_i^{price} and promotional lift β_i^{promo} coefficients is shown in Figure 1.

An obvious deficiency of the log-linear model is its implicit assumption of non-zero demand in each observation, which limits its application to either fast-moving items that always have positive weekly demand or items aggregated to higher levels of the location hierarchy such pricing-zone store clusters. Section 2.1.5 gives modeling alternatives that allow for slow-moving items.

2.1.3 The cannibalization effect and its equivalence to the MNL model

Define the utility of item i at time t as

$$u_i(p_{it}, r_{it}) = \exp(\beta_i^0 + \beta_i^{price} p_{it} + \beta_i^{promo} r_{it}) \quad (2)$$

Given a category C of similar items, define the total category utility $z(t)$ at time t as

$$z(t) = \sum_{i \in C} u_i(p_{it}, r_{it})$$

and let $\bar{z}$ be the average of $z(t)$ over the entire observation period. Finally, we define the cannibalization variable as

$$b_i(t) = \frac{z(t) - \bar{z}}{\bar{z}} \tag{3}$$

That is, the cannibalization variable is defined as the relative deviation of the total category utility at a specific time period from the average category utility over the entire observation period. The intuition behind this definition is that when the total utility of the category increases, for example, due to price discounts or promotions, the demand for a specific item will be cannibalized by other items unless the item's utility is increased also. Following this definition, the value of the cannibalization coefficient should satisfy

$$-1 \leq \beta_i^{cann} \leq 0 \tag{4}$$

Below, we show that this model of cannibalization is asymptotically equivalent to the commonly used Multinominal Logit model (see e.g. [RT04]), which can be represented in the form:

$$D_i^{MNL}(t) = N \frac{U_i^{MNL}(p_{it})}{1 + \sum_{j \in C} U_j^{MNL}(p_{it})} \tag{5}$$

where N is the market size and the unit term in the denominator represents the no-purchase option.

In order to see the equivalence of our model to (5), note the asymptotic equality

$$e^x \approx \frac{1}{1 - x} \tag{6}$$

for small x and rewrite the model equation (1) exponentiating both parts and using expressions for utility (2) and the cannibalization variable (4):

$$d_{it} = s_i(t) \frac{u_i(p_{it}, r_{it})}{1 - \beta_i^{cann} \frac{z(t) - \bar{z}}{\bar{z}}} \tag{7}$$

After simple algebraic manipulations, the formula (7) can be rewritten as

$$d_{it} = s_i(t) \frac{\bar{z}}{-\beta_i^{cann}} \cdot \frac{\kappa u_i(p_{it}, r_{it})}{1 + \sum_{j \in C} \kappa u_j(p_{it}, r_{it})}$$

where $\kappa = \frac{\beta_i^{cann}}{\beta_i^{cann} + 1} \cdot \frac{1}{\bar{z}}$. By setting the MNL utility as

$$U_i^{MNL}(p_{it}) = \kappa u_i(p_{it}, r_{it})$$

and market size

$$N = s_i(t)\frac{\bar{z}}{\beta_i^{cann}}$$

we obtain an expression equivalent to (5) by noting that coefficient κ can be absorbed into the linear function of price.

By further analyzing (7), we can see that when $\beta_i^{cann} = 0$ there is no cannibalization of the demand, as the entire demand increase for a specific item is drawn from the no-purchase option. On the other hand, when $\beta_i^{cann} = -1$, the entire demand is cannibalized. That is, any demand increase for a given item leads to demand decrease for all other items in the category and the total demand for the category stays constant. For most of the retail categories encountered in practice, the estimated value of β_i^{cann} is in the range of [-0.9, -0.7].

2.1.4 The Expectation Maximization (EM) Approach to Estimating Demand Cannibalization

Since the cannibalization variables cannot be observed or computed without the values of the model parameters, we propose the following two-step iterative EM procedure for estimation:

1. Given the current values of the demand-model parameters $\vec{\beta}$ and the values of the directly observed variables, compute the cannibalization variable values $b_i(t)$ for each time period using formula (3).

2. Re-estimate the parameters $\vec{\beta}$.

To carry out the Step (1) of the above EM procedure, use initial guesses for the parameters based on prior knowledge of the category. The specific details of our implementation are provided in section 5

2.1.5 Demand modeling for slow-moving items

As mentioned above, the log-linear model is not applicable when there are variables with significant numbers of 0 values. Two alternative approaches are: demand smoothing and Poisson regression.

We apply demand smoothing in a similar fashion to the well-known additive smoothing (e.g. [CG99]): Suppose there are n zeros among the demand observations and the total demand for the observation period is D. Then we transform the observed demand according to:

$$d_t := \begin{cases} d_t(1 - \frac{n}{D}\varepsilon), & \text{if } d_t > 0. \\ \varepsilon, & \text{otherwise.} \end{cases}$$

It is easy to see that the total demand stays the same and non-zero demand values are decreased by the same factor. The parameter ε can be thought of as a regularization parameter. When $\varepsilon = \frac{D}{n+D}$, zero and unit demands are transformed into the same value.

Poisson regression provides a more accurate way of predicting events with low counts than fitting the log of the number of the events via OLS (see [OK10]). In addition, the number of units sold in a week can be 0. The main disadvantage of Poisson regression is that its maximum-likelihood estimates lack a closed-form expression and must be found by numerical methods. However, its log-likelihood function is concave, which allows finding its maximum by gradient search. Following the common procedure for Poisson regression, we model the demand for product i at time t as Poisson distributed, and so the probability density is given by

$$p(d_i(t)|x_i(t);\vec{\beta_i}) = \frac{\lambda_i(t)^{d_i(t)}}{d_i(t)!}e^{-\lambda_i(t)}$$

where $x_i(t)$ and $\vec{\beta_i}$ are, respectively, the vectors of independent variables and estimated coefficients in (1) and
$\lambda_i(t) = E(d_i(t)|x_i(t)) = e^{\vec{\beta_i}' x_i(t)}$ is the expected value of demand for the variable vector $x_i(t)$.

Given a series of T observations $(d_i(t), x_i(t))$, $t = 1, \ldots, T$, the logarithm of the likelihood becomes:

$$l(\vec{\beta_i}) = \sum_{t=1}^{T}(d_i(t)\vec{\beta_i}' x_i(t) - e^{\vec{\beta_i}' x_i(t)} - \log(d_i(t)!)) \tag{8}$$

Since the last term in the sum in (8) does not depend on $\vec{\beta_i}$, it can be dropped from the expression to be maximized. Then after adding a L_2 regularization term, the problem of finding the log-likelihood estimator of the parameter becomes:

$$\max_{\vec{\beta_i}} \sum_{t=1}^{T}(d_i(t)\vec{\beta_i}' x_i(t) - e^{\vec{\beta_i}' x_i(t)}) - \frac{T}{2}\alpha\|\vec{\beta_i}\|_2^2 \tag{9}$$

where α is the regularization penalty parameter estimated via cross-validation.

2.2 Hierarchical Estimation

This hierarchical procedure mitigates the potentially insufficient statistical power which results from having a limited number of observations for a single item.

2.2.1 Bottom-up pooling stage

At this stage, we pool together multiple similar items to estimate their common parameters. In model (1) we can expect similar products to have the same promotion lifts, the same cannibalization effect, and the same seasonality. However, despite the similarity between the items, their prices and demand volume may be quite different, which leads to potentially large difference for β_i^{price} and β_i^0 coefficients. An example of the latter could be the same item sold at two different locations with different market size. An example of the former could

be selling different package sizes of the same product while demand for it is counted only as units sold. To alleviate this problem, we modify the model (1) by scaling demand and price by their averages. We replace demand with $\tilde{d}_i(t) = \frac{d_i(t)}{\bar{d}_i}$ and price with $\tilde{p}_i(t) = \frac{p_i(t)}{\bar{p}_i}$, where $\bar{d}_i$ is the geometric mean and $\bar{d}_i$ is the arithmetic mean of the regular, i.e. non-promotion, demand and prices, respectively. Thus model (1) becomes:

$$\begin{aligned}\log \tilde{d}_i(t) = \beta_i^0 + \beta_i^{price}\tilde{p}_i(t) + \beta_i^{promo} r_i(t) + \beta_i^{cann} b_i(t) + \\ \beta_{it}^{hol} I_{W_{hol}}(t) + \beta_i^{sin}\sin(\frac{2\pi}{52}t) + \beta_i^{cos}\cos(\frac{2\pi}{52}t)\end{aligned} \tag{10}$$

Since the output variable in model (10) is now centered at zero, the β_i^0 coefficient vanishes for all items. On the other hand, following the commonly used definition for price elasticity of demand (PED) as $E(p) = \frac{\partial d}{\partial p} \cdot \frac{p}{d}$, the β_i^{price} coefficient in (10) can now be interpreted as PED at the average price $\bar{p}_i$. As it is common in practice to have similar products priced at approximately the same PED, we can pool together similar products and fit the model (10) using all observations with $i \in S$ and $t \in T$ where S is the set of all similar items from multiple locations and merchandise groups.

2.2.2 Top-down hierarchical fitting stage

At this stage, the model is first fit at a high level of the merchandise and location hierarchies using a large set of items and the full set of features. Then at the lower levels of the hierarchies the model is fit using subsets of items and features. We obtain the vector $\hat{\beta}$ of estimated coefficients in (10) based on all items in S by using the following ridge regression approach:

$$\hat{\beta} = \arg\min_{\vec{\beta}}(\|\mathbf{y} - \vec{\beta}\mathbf{X}\|_2^2 + \frac{T|S|}{2}\alpha\|\vec{\beta}\|_2^2)$$

where $\mathbf{y}$ is the vector of the output variable observations $y_{it} = \log d_i(t), i \in S, t = 1, \ldots, T$, and $\mathbf{X}$ is the matrix of corresponding variables. α is the penalty parameter estimated via cross-validation.

In order to estimate the individual item coefficients $\hat{\beta}_i$ and counter any singularity in the individual-item matrix of observations $\mathbf{X}_i$, we regularize the model for an individual item by adding the L_2 norm of the deviation of the coefficient vector $\vec{\beta_i}$ from the group-based vector $\hat{\beta}$, with α_i being the penalty on the deviation:

$$\hat{\beta}_i = \arg\min_{\vec{\beta_i}}(\|\mathbf{y} - \vec{\beta_i}\mathbf{X}\|_2^2 + \frac{T}{2}\alpha_i\|\vec{\beta_i} - \hat{\beta}\|_2^2) \tag{11}$$

Equivalently, denote the deviation of the coefficients as $\Delta\vec{\beta_i} = \vec{\beta_i} - \hat{\beta}$ and the vector of residuals as $\mathbf{e}_i = \mathbf{y}_i - \hat{\beta}\mathbf{X}_i$. Expressing the first term in (11) as $\|\mathbf{y}_i - \vec{\beta_i}\mathbf{X}_i\|_2^2 = \|(\mathbf{y}_i - \hat{\beta}\mathbf{X}_i) - (\vec{\beta_i} - \hat{\beta})\mathbf{X}_i)\|_2^2$ gives:

$$\hat{\beta}_i = \hat{\beta} + \arg\min_{\Delta\vec{\beta_i}}(\|\mathbf{e}_i - \Delta\vec{\beta_i}\mathbf{X}_i\|_2^2 + \frac{|T|}{2}\alpha\|\Delta\vec{\beta_i}\|_2^2) \tag{12}$$

This is essentially a ridge regression.

To have certain parameters, e.g. seasonality and cannibalization, estimated only at the group level and fixed for an individual item, use only a sub-vector of $\Delta\vec{\beta}_i$ of (12) together with the corresponding submatrix of $\mathbf{X}_i$. This is also equivalent to penalizing the dropped components with infinite penalty.

This approach can be easily generalized to several levels of merchandise and location hierarchies and to using other regression models, e.g. the regularized Poisson regression shown in (9). We provide the specifics of our implementation in section 5.

3 The Markdown Optimization Problem with Price-Elasticity Differentiation and Multiple Fulfillment Centers

Following are several formulations of the markdown optimization problem at different levels of generality.

First, we introduce the basic markdown problem assuming a single customer group served by a single fulfillment center (FC). Next, we extend the formulation to consider multiple customer groups. The final formulation has multiple FCs incurring different shipping costs. For each formulation, we describe the closed-form solution or, if it does not exist, the algorithm to obtain the optimal or near-optimal solution. The scope of this paper is limited to markdown optimization for a single merchandise item. Another formulation of the markdown problem with multiple items and finite price set is considered in the forthcoming paper [LJUV].

3.1 The Basic Markdown Optimization Problem: A Single Customer Group Served From One FC

Suppose an item has inventory of S units that must be sold within time horizon T, and that the item is not replenished. Given the demand $d_t(p_t)$ as a function of price p_t at time period t, the basic markdown problem is to find the prices which maximize the total revenue $\sum_{t=1}^{T} p_t s_t$ such that the sales s_t at time t does not exceed the demand at that time and the total amount sold does not exceed S:

$$\max_{p_t, t=1,\ldots,T} \sum_{t=1}^{T} p_t s_t \tag{13}$$

subject to:

$$\forall t = 1, \ldots, T : s_t \leq d_t(p_t) \tag{14}$$

$$\sum_{t=1}^{T} s_t \leq S \tag{15}$$

It is easy to see that in the above optimization problem the constraint (14) is a strict inequality for at least one period t only if there is not enough inventory. In this case, under the assumption of continuity of price variables and demand functions, it would be possible to increase revenue by slightly increasing the price while still clearing all inventory. Thus, when prices and the demand function are continuous, the sales in the optimal solution should be equal to the demand in each t. Therefore, in the formulation (13)-(15) the sales variable can be replaced by demand. By using the expression for the demand from model (1) and omitting item index i, the markdown optimization problem formulation can be written as:

$$\max_{p_t, t=1,\ldots,T} \sum_{t=1}^{T} p_t a_t e^{\beta^{price} p_t} \tag{16}$$

subject to:

$$\sum_{t=1}^{T} a_t e^{\beta^{price} p_t} \leq S \tag{17}$$

where all demand factors other than price are absorbed into the parameter a_t.

Theorem 1. *In the optimal solution to problem* (16)-(17)*, all prices* p_t *are the same.*

Proof. The optimal solution to the unconstrained problem (16), found by solving for p_t in $\frac{\partial d_t(p_t)}{\partial p_t} = 0$, gives a constant p_t:

$$p_t^* = p^* = -\frac{1}{\beta^{price}} \tag{18}$$

Since in almost all practical applications the demand is a strictly decreasing function of price, $\beta^{price} < 0$ and so p^* is positive. If the constraint (17) is satisfied for $p_t = p^*$, then all prices are the same. Otherwise, the constraint (17) becomes an equality. Following the general framework laid out in Chapter 5 of [RT04], we express the price as the function of demand, namely, $p_t(d_t) = \frac{1}{\beta^{price}} \log \frac{d_t}{a_t}$, which is possible due to strict monotonicity of the demand function. Then the problem (16)-(17) can be rewritten as:

$$\max_{d_t, t=1,\ldots,T} \frac{1}{\beta^{price}} \sum_{t=1}^{T} (\log d_t - \log a_t) d_t$$

subject to:

$$\sum_{t=1}^{T} d_t = S$$

By the orthogonality of the objective function gradient to the constraint function in the optimal solution, we obtain $\frac{\partial}{\partial d_t}(d_t(\log d_t - \log a_t)) = k$, or $d_t = a_t e^{k-1}$, where k is a constant obtained by substituting the expression for d_t into constraint (17). Finally, the expression for the optimal inventory clearing price becomes: $p_t = p^{ICP} = \frac{1}{\beta^{price}} \log \frac{S}{\sum_{t=1}^{T} a_t}$, which is the same for each period. □

Corollary 1.1. *The optimal solution to* (16)-(17) *is* $p_t = p^{OPT} = \max(p^*, p^{ICP})$, *or*

$$p_t = p^{OPT} = \max(-\frac{1}{\beta^{price}}, \frac{1}{\beta^{price}} \log \frac{S}{\sum_{t=1}^{T} a_t}) \tag{19}$$

3.2 Markdown Optimization for Multiple Customer Groups Served From the Same FC

Denote the cost of serving each customer group ℓ as c_ℓ. The markdown problem is now to maximize profit:

$$\max_{p_\ell, \ell \in L} \sum_{t=1}^{T} \sum_{\ell \in L} (p_\ell - c_\ell) a_{\ell t} e^{\beta_\ell^{price} p_\ell} \tag{20}$$

subject to:

$$\sum_{t=1}^{T} \sum_{\ell \in L} a_{\ell t} e^{\beta_\ell^{price} p_\ell} \leq S \tag{21}$$

where $a_{\ell t}$ is the multiplicative demand factor for group ℓ at time t.

The main observation here is that if the overall inventory level is sufficiently high so that constraint (21) is not binding, then the solution to the optimization problem is the profit-maximizing price for each customer group $p_\ell^{OPT} = p_\ell^* = -\frac{1}{\beta_\ell} + c_\ell$. Otherwise, the optimal price is computed as the inventory-clearing price for each individual group for a certain amount of allocated inventory s_ℓ. Indeed, suppose there are some groups with profit-maximizing price and others with inventory-clearing price. Then by reallocating the excess inventory from the former groups to the latter we can either make all prices profit-maximizing or make the constraint (21) tight. In the latter case, with (21) tight, solution is similar to the solution for (16)-(17). Then the ICP for customer group ℓ is $p_\ell^{ICP} = \frac{1}{\beta_\ell^{price}} \log \frac{s_\ell}{\sum_{t=1}^{T} a_t}$ and the the optimization problem becomes

$$\max_{p_t, t=1,\ldots,T} \sum_{\ell \in L} (\frac{1}{\beta_\ell^{price}} \log \frac{s_\ell}{\sum_{t=1}^{T} a_t} - c_\ell) s_\ell$$

subject to:

$$\sum_{\ell \in L} s_\ell = S$$

By solving equation:

$$\frac{\partial}{\partial s_\ell} ((\frac{1}{\beta_\ell^{price}} \log \frac{s_\ell}{\sum_{t=1}^{T} a_{\ell t}} - c_\ell) s_\ell) = k$$

we obtain

$$s_\ell = \sum_{t=1}^{T} a_{\ell t} e^{\beta_\ell^{price} k - 1 + \beta_\ell^{price} c_\ell}$$

After that, the constant k can be found from equation (21). The above results can be summarized as the following theorem:

Theorem 2. *The solution to the markdown optimization problem* (20)-(21) *can be found as:*

$$p_\ell^{OPT} = \begin{cases} -\frac{1}{\beta_\ell^{price}} + c_\ell, & \text{if } \sum_{\ell \in L} e^{\beta_\ell^{price} c_\ell - 1} \geq \frac{S}{\sum_{t=1}^{T} a_t}; \\ -\frac{1}{\beta_\ell^{price}} + c_\ell + k, & \text{otherwise.} \end{cases} \tag{22}$$

where k is obtained as a solution to

$$\sum_{\ell \in L} e^{\beta_\ell^{price}(k + c_\ell)} = e \frac{S}{\sum_{t=1}^{T} a_t} \tag{23}$$

Note that in the case of the inventory clearing prices in equation (22), the optimal prices for each group are obtained by increasing their profit maximization prices by the same constant, which is found by numerically solving the equation (23). Despite the fact that this equation does not have a closed form solution, it is computationally easy to find it by applying a one-dimensional binary search.

3.3 Markdown Optimization for Multiple Customer Groups Served From Multiple FCs

In most practical applications, an item with a limited life span is in relatively few FCs, in fact rarely exceeding ten. On the other hand, the number of customer groups varies from relatively small to more than 100, especially when determined by the geographic locations. We formulate the markdown optimization problem as a non-linear integer programming problem and propose a fast algorithm to obtain a near-optimal solution in the case of a large number of customer groups.

3.3.1 The non-linear integer program

Suppose there are M FCs serving L customer groups that may be defined either by their locations or other parameters. At each FC m, there is initial product inventory S_m. The product has remaining life time T. Given the product demand function $d_\ell(p)$ at customer group ℓ, and unit delivery cost to the customer group ℓ from FC m as $c_{m\ell}$, the optimization problem is to find the customer-group-specific prices p_ℓ and amount of inventory $x_{m\ell}$ delivered from m to ℓ that maximizes profit, subject to FC inventory constraints.

The total amount of inventory allocated to customer group ℓ is $\sum_{m \in M} x_{m\ell}$ and the optimal markdown price for that group is $p_\ell^{OPT} = -\frac{1}{\beta_\ell^{price}} \max(1, -\log \frac{\sum_{m \in M} x_{m\ell}}{\sum_{t=1}^{T} a_{\ell t}})$ according to equation (19). The entire

formulation then becomes:

$$\max \sum_{\ell \in L} s_\ell (p_\ell - \sum_{m \in M} c_{m\ell} x_{m\ell}) \tag{24}$$

subject to:

$$\sum_{\ell \in L} x_{m\ell} \leq S_m \tag{25}$$

$$s_\ell = \sum_{m \in M} x_{m\ell} \tag{26}$$

$$s_\ell \leq e \sum_{t=1}^{T} a_{\ell t} \tag{27}$$

$$p_\ell = \frac{1}{\beta_\ell^{price}} \log \frac{s_\ell}{\sum_{t=1}^{T} a_{\ell t}} \tag{28}$$

Since the above optimization problem (24)-(28) is essentially non-linear, we apply a Randomized Decomposition approach to obtain a near-optimal solution, which is described in section 4.

3.3.2 Approximate solution in case of a large number of customer groups

Since the solution approach to optimization problem (24)-(28) does not scale well when the number of customer groups becomes large, we propose an approximate solution by noticing that in this case each customer group is supplied from only one FC as the demand per group becomes very low. Let zero-one variable $y_{m\ell}$ denote the selection of the FC m to supply customer group ℓ. By assuming that each customer group is supplied from a single FC, the variables $y_{m\ell}$ satisfy $\sum_{m \in M} y_{m\ell} = 1 \forall \ell \in L$ which determines partitioning of the set L into $|M|$ subsets, $L_1, \ldots, L_{|M|}$, corresponding to the supplying FCs. After that, for the given partition, the markdown optimization problem can be decomposed into solving optimization problem (20)-(21) for each m with $S = S_m$. We denote the optimal solution to this problem as MDO_m, which is a function of variables $y_{m\ell}$. Then the problem can be formulated as

$$\max \sum_{m \in M} MDO_m(y_{m\ell}) \tag{29}$$

subject to:

$$\forall \ell \in L : \sum_{m \in M} y_{m\ell} = 1 \tag{30}$$

$$\forall \ell \in L, \ \forall m \in M : y_{m\ell} \in \{0, 1\} \tag{31}$$

The optimization problem (29)-(31) is solved by the Randomized Decomposition framework.

4 The Randomized Decomposition Framework

In this section we describe the Randomized Decomposition Framework (RDF) for solving non-linear, non-convex mathematical programs. The framework implementation, called RDSolver, is implemented in Java, and is currently being shipped with Oracle products for project portfolio optimization and vehicle routing. It has also been integrated into additional Oracle products for discount optimization and assortment optimization and is under evaluation for use in other areas. The details of the algorithm are presented in our recent paper [MRW18].

RDF, presented in Algorithm 1, combines a large neighborhood search (LNS) metaheuristic for exploring the solution space using an extension of multi-block block coordinate descent (BCD) called the Random Sampler Cyclic BCD (RSC-BCD) combined with a solution perturbation method called RDPerturb to escape local optima.

Algorithm 1 RDF

1: Construct an initial feasible solution $\hat{x}$.
2: $x_{\text{best}} \leftarrow$ RSC-BCD($\hat{x}$)
3: **while** termination criteria not satisfied[1]
4: $\quad \{x_{\text{candidate}},\} \leftarrow$ RD-Perturb(x_{best})
5: $\quad x^* \leftarrow$ RDS-BCD($x_{\text{candidate}}$)
6: $\quad$ **if** $cost(x^*) < cost(x_{best})$ **then**
7: $\quad\quad x_{\text{best}} \leftarrow x^*$
8: **end while**
9: **return** x_{best}

[1] Max run time, reaching a "good enough solution", inability to improve the solution for a long time, etc.

4.1 RSC-BCD

Block coordinate descent methods (BCD) [Wri15] are well suited for large-scale unconstrained optimization problems, but they lack the ability to target complex constraints efficiently. RDS-BCD mitigates the problem by "randomly" creating blocks at the beginning of each iteration, and solving the blocks in sequential order once per iteration. The variables not assigned to the block being optimized are considered as constants.

RSC-BCD pseudo code is shown in Algorithm 2. The outer while loop describes the cyclic process of creating and solving blocks (i.e. sub-problems). The number of iterations is controlled by an input parameter *nTrials*, which defines how many times the algorithm will seek a better solution without any improvement in the value of the objective (cost) function.

Grouping of the variables is done by randomly selecting them without replacement (lines 6-14), with higher probability given to variables joined by active constraints. Creating block sets by choosing variables at random is simple, and is efficient for unconstrained problems, but it is sub-optimal for constrained problems. The way we chose to deal with this limitation is described in lines 9-13. To allow the selected variable x_y to change its value (if it is an operand

Algorithm 2 RSC-BCD

Require: Maximum number of trials with no improvement, *nTrials*
Min sub-problem size, k, problem size (number of decision variables), n
X := {decision variables $x_i : i \in \mathcal{Z}, 1 < i \leq n$}; C := {constraint functions}

$x \leftarrow \hat{x}$; *cnt*$\leftarrow 0$
while *cnt* $<$ *nTrials*
  $j \leftarrow 1$ *bf* $\leftarrow$ *false*
  while $\bigcup_{i=1}^{j} \mathrm{S}_i \neq \mathrm{X}$
    $\mathrm{S}_j \leftarrow \varnothing$
    while $|\ \mathrm{S}_j| < k$
      Select x_y randomly from $\mathrm{X} \backslash \bigcup_{i=1}^{j} \mathrm{S}_i$
      $\mathrm{S}_j \leftarrow \mathrm{S}_j \cup \{x_y\}$
      for each active constraint $c_k \in C$ where x_y is an operand **do**
        T := {decision variables that are operands of c_k}
        Select x_z from $\mathrm{T} \backslash \mathrm{S}_j$
        $\mathrm{S}_j \leftarrow \mathrm{S}_j \cup \{x_z\}$
      end for each
    end while
    Create a sub-problem from the original problem by considering variables
    $x_i \notin \mathrm{S}_j$ as constants. Find $x_i^* \in S_j$, that optimize *cost*(x)
    $x \leftarrow x^*$.
    if *cost*$(x^*) <$ *cost*(x) **then** *cnt* $\leftarrow 0$
  end while
end while
return x

of an active constraint), we choose a random variable (not already being chosen by any other block) from all its active constraints and add it to the block. This allows us to change the value of x_y by also changing the values of variables in all its active constraints. This approach can be modified if we are using domain knowledge to better group variables. At least one variable is chosen at random, and the rest of variables are then added in a semi-random fashion based on domain knowledge. The variable selection can be specialized to give preference to variables of larger (or smaller) values, or variables that are grouped around some value, or variables that have wide variations, and so forth. The limitation of the approach is that in the worst case scenario (constraints are all linked by a single variable) we may end up with a block as large as the original problem. We are currently working on elimination of this issue.

4.2 RD-Perturb

The LNS metaheuristic, RD-Perturb, is similar to many LNS algorithms in the sense that it attempts to escape a local optimum by changing a large number of the current decision variable values. The decision variables to be changed are selected at random, but the selected random decision variable values are required to remain in the set of feasible solutions, unlike other LNS algorithms that may select a completely random and possibly infeasible starting point.

RD-Perturb explores the search space with a bias towards feasible solutions near the current best solution. It randomly chooses some or all variables, and changes their values to a new starting point that is within some ϵ of the current

Algorithm 3 RD-Perturb

Require: Starting (feasible) solution x_{start},constraints
Distribution parameters: $p_{\min}, p_{\max}, \lambda$
X := {set of decision variables $x_i : i \in \mathcal{Z}, 1 < i \leq n$}
Draw random number m from $\lfloor \text{f}(y|ps_{min}, ps_{max}, \lambda) \rceil$ // Equation 32
V := $\{x_i \in \text{X}: i \in Y, Y \sim \mathcal{U}(0, n)\}$, $|\text{V}| \leq m$
F := {feasible solutions in proximity of x_{start}}
Select $x_{\text{candidate}}$ randomly from F
return $\{x_{\text{candidate}}\}$

result.

Note that the set V in Algorithm 3 can be also constructed using an algorithm based on domain knowledge.

The number of variables m is randomly chosen from a distribution. Any distribution could be used, but empirically we found that the best was the truncated exponential with density function

$$f(y|p_{min}, p_{max}, \lambda) = \frac{\lambda e^{-\lambda y}}{e^{-\lambda ps_{min}} - e^{-\lambda ps_{max}}} \tag{32}$$

where $1 \leq ps_{min} < ps_{max} \leq n$, and n is the total number of decision variables (problem size). We round to the nearest integer since m is integer.

5 Computational experiments

5.1 Data Description

We applied our predictive and prescriptive models to test data reflecting the sales of about 30 items constituting the category of USB flash drives (a.k.a. "memory sticks"). We assumed that the items in the category vary in their storage capacity, brands, and other features but otherwise functionally substitutable. The sales data contained two years of online order transactions, which were aggregated over 100 customer locations corresponding to the first two digits of the order destination zip codes. The costs were the UPS shipment rates [UPS].

5.2 Predictive Model Experiment

The hierarchical demand model described in section 2 was implemented at three levels of hierarchy: electronics department, flash drive category, and individual items. We selected five holiday weeks with distinctly higher demand: two back-to-school weeks, the "Cyber Monday" week, and two weeks of Christmas sales. Once the seasonality coefficients were estimated at the department level, they were fixed for all lower levels. The cannibalization and promotion lift coefficients were estimated at the category level and then kept fixed for the individual items. The value of the cannibalization coefficient was estimated at -0.8 and promotion lift coefficient was estimated to be 0.9 corresponding to about 2.5 increase of

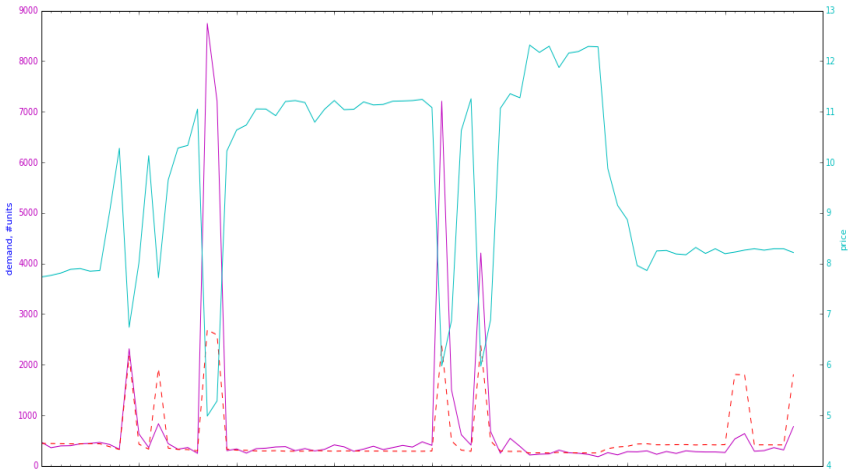


Figure 2: Results of fitting the demand model at the category level. The average price for the category is shown in cyan. The demand is shown in red with out-of-sample prediction shown by dashed line

sales during promotions. The price elasticity coefficient was estimated to be about -1.8 at the category level implying that the category products are moderately elastic. Finally, the price elasticity for the individual items was estimated by increasing the regularization coefficient high enough if necessary to ensure that the correction to the item price sensitivity would keep it elastic. Since the sales signal for the specific locations was too weak for an individual items, we consider the variation of the average price elasticity of demand at the department level for each location and adjusted each item price elasticity at the location level accordingly. The results of fitting the demand at the category level are shown in Figure 2.

The results of fitting the demand model at the location level for a particular product item are shown in Figures 3 and 4 demonstrating how price elasticity and $\bar{a}_\ell = \frac{1}{T}\sum_{t=1}^{T} a_{\ell t}$ coefficients vary among the locations. Note that in Figure 3 the areas with higher income level have lower price elasticity. Also, it can be seen from Figure 4 that $\bar{a}_\ell$ estimates are generally higher for more populated locations.

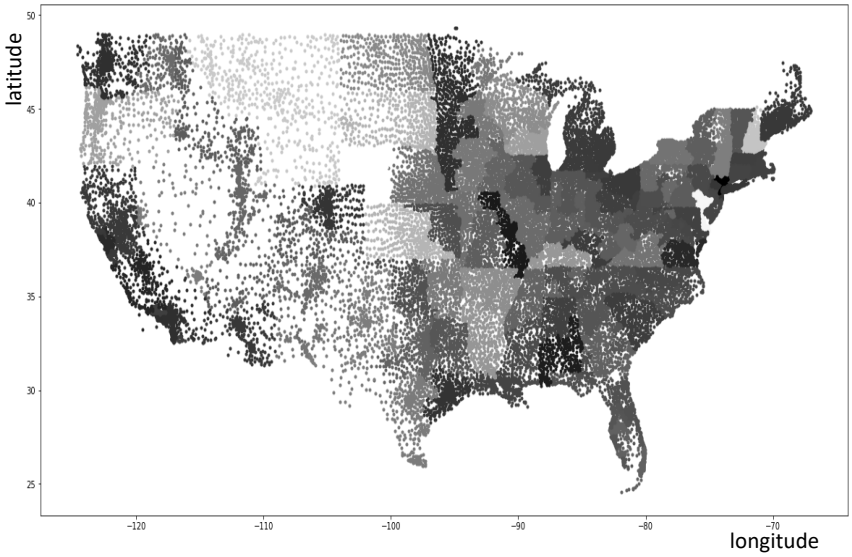


Figure 3: Range of the estimated β_ℓ^{price} coefficients shown by grayscale range from dark (lowest elasticity) to light gray (highest elasticity)

5.3 Prescriptive Model Experiment

For evaluating the prescriptive model, we solved the markdown optimization problem for about 100 location-based customer groups supplied from two FCs located in Los Angeles, CA, and Atlanta, GA, areas as shown in Figure 5. Each FC has an excess of inventory that must be sold within the next 13 weeks. All inventory that is not sold within this period is salvaged through the secondary channels at near-zero price. We assume that in order to boost the sales the retailer would promote the item by increasing its online visibility. On the other hand, we assume that the entire item inventory is low enough so that the optimal discount is inventory clearing rather than the profit-maximizing price as explained in 3.3. Since the number of customer groups is large, we use the approach described in section 3.3.2. Finally, our solution to the problem is shown by two color schemes shown in Figure 5: The one ranging from green to red show the location groups served form the Los Angeles FC with lower prices shown in green and higher shown in red. The other ranging from cyan to purple shows the location groups served from the Atlanta FC with lower prices shown in blue and higher prices shown in purple. Overall, it can be seen that our recommended prices depend on both proximity to the serving FC and the overall price sensitivity of the group. For example, in the areas around Los Angeles, the item is sold at lower price despite its relatively low elasticity because of their proximity

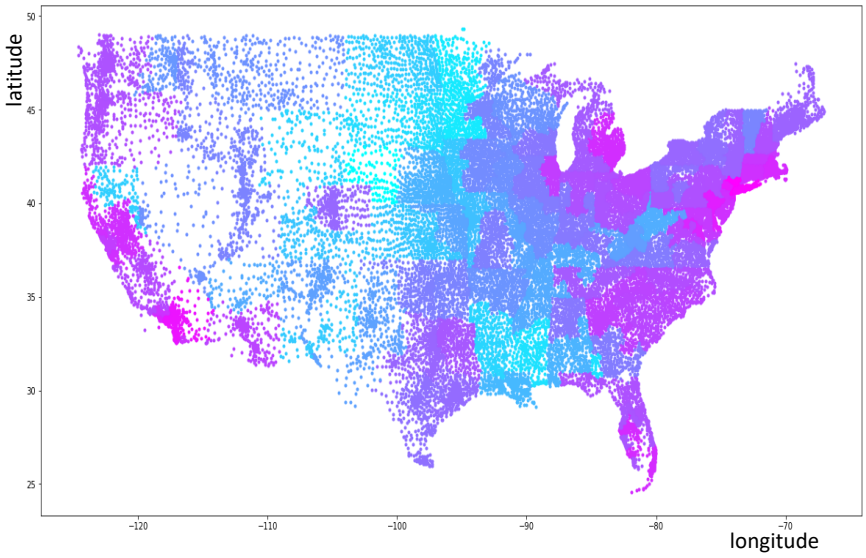


Figure 4: Range of the estimated $\bar{a}_\ell$ coefficients shown by color gradation from cyan (low) to purple (high)

to the FC and low delivery cost. On the other hand, the generally more price sensitive states of Alabama and Mississippi are assigned lower price than less price sensitive Georgia despite the proximity of the latter to the respective FC.

In order to illustrate the advantage of our approach we computed the relative increase in profit using our diversified-price solution as compared to the optimal solution with a single price as described in 3.1. We found the projected increase in revenue to be about 2%, which assuming the original profit margin of about 5% translates into almost 60% increase of the profit margin when price diversification is applied. However, due to the demand cannibalization effect, the demand for other items became slightly lower, which resulted in overall increase in profit margin to be about 14%.

6 Summary and Future Work

This paper presents a novel approach to markdown optimization as faced by an e-commerce retailer fulfilling orders from geographically dispersed FCs to multiple customer groups that differ by their price sensitivity and cost of fulfillment. We formulated a comprehensive predictive–prescriptive analytics approach to finding a profit maximizing delivery assignment and pricing solution based on the observed sales history. We also conducted empirical evaluations based on

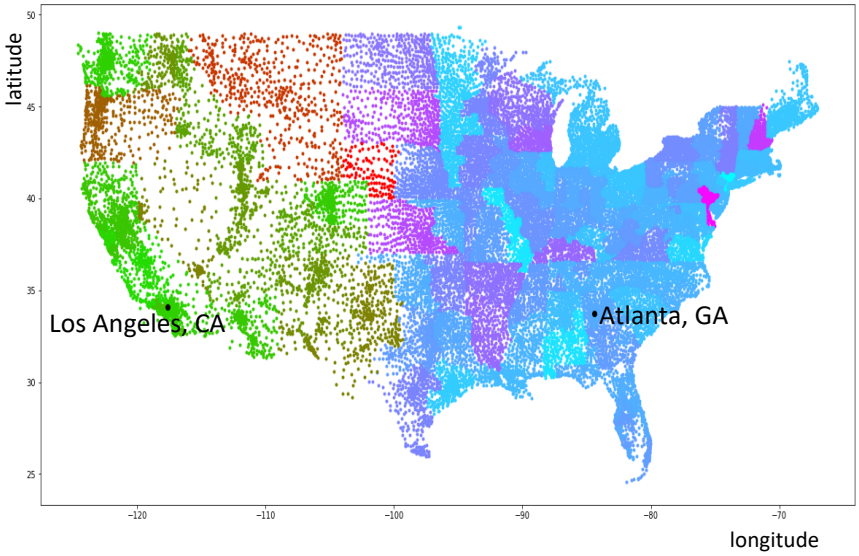


Figure 5: Assignments and pricing in case of two FCs: Atlanta and Los Angeles. Prices of products assigned to Atlanta are shown in color scale from cyan (low prices) to purple (high prices). Los Angeles assignments are ranging from green (low prices) to red (high prices)

the data mimicking performance of a real e-commerce retailer, which showed that implementing this approach results in significant increase of the retailer profit margin. As the scope of our optimization model is limited to optimizing markdown price for a single product while accounting for the demand cannibalization of the other products, one of the potential directions for the future work would be to develop an approach to solving the problem of simultaneous markdown optimization for multiple products.

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